

Phase 8
Project Demonstration Phase

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The demonstration video provides a complete technical walkthrough of the AI BlogNest API, covering the codebase architecture, server initialization, and live endpoint verification.

- **Project File Explanation** – Walks through the VS Code workspace and src directory following the MVC pattern, explaining the database setup (config/db.js), data schemas (models), business logic (controllers), API route mappings (routes), JWT security and error handling (middleware), and the Google Gemini AI integration (services/geminiService.js).
- **Backend Execution** – Demonstrates configuring environment variables (.env), initializing the Express application (app.js and server.js), and starting the local backend server via npm start alongside an active MongoDB connection.
- **API Endpoint Testing** – Uses Thunder Client / Postman to showcase the exact request payloads, authorization headers, HTTP status codes, and JSON responses across all six core endpoints:
 - **User Registration (POST /api/auth/register)**: Sends a raw JSON body containing name, email, and password. The server hashes the password, creates the user record in MongoDB, and returns a 201 Created status with the generated user ID and profile details.
 - **User Login (POST /api/auth/login)**: Submits the registered email and password credentials. The backend validates the account and returns a 200 OK status along with a signed, stateless JWT token used for accessing protected routes.

- **Create Blog (POST /api/blogs):** Passes the JWT Bearer token in the Authorization header alongside a JSON payload with title, content, and category. The server links the post to the logged-in author and returns a 201 Created status with the saved blog document and timestamps.
- **Fetch Blogs (GET /api/blogs):** Queries the database for all stored posts and returns a 200 OK status with a JSON array of blog documents, automatically populated with the author's ID, name, and email.
- **Generate AI Blog (POST /api/ai/generate-blog):** Sends a JSON prompt containing a topic and category. The controller calls the Gemini AI service to synthesize a full-length, structured blog post, saves it directly to MongoDB under the authenticated author, and returns a 201 Created status.
- **Summarize Blogs (POST /api/ai/summarize):** Submits content text in the request body. The AI engine condenses the input and returns a 200 OK status with a concise summary abstract.

Demo Video: [Demo Video Link](#)
